$$\mathbf{x}_t^{up} = W\mathbf{x}_t + \mathbf{b} \tag{5}$$

where $W \in \mathbb{R}^{4\times 2}$ and $\mathbf{b} \in \mathbb{R}^4$ are learnable parameters.

4.2 Adaptive Weighting Mechanism

To dynamically balance the contributions of the global and local branches, we introduce a learnable, timestep-conditioned weighting mechanism:

$$\mathbf{w}(t) = \text{Softmax}(\text{MLP}_w(t)) \tag{6}$$

where $\mathbf{w}(t) \in \mathbb{R}^2$ represents the weights for the global and local branches at timestep t. The weight network MLP_w is implemented as:

$$\text{MLP}_w(t) = \text{Linear}_2(\text{LeakyReLU}(\text{Linear}_1(\text{SinusoidalEmbedding}(t)))) \tag{7}$$

This design allows for complex weight computations, enabling nuanced adaptations of the global-local feature balance across different timesteps. The use of LeakyReLU activation and multiple linear layers provides the network with the capacity to learn non-linear relationships between the timestep and the optimal feature balance.

4.3 Combined Denoising Process

The final denoising prediction is a weighted combination of the global and local branch outputs:

$$\epsilon_\theta(\mathbf{x}_t, t) = w_1(t) \cdot \epsilon_\theta^{\text{global}}(\mathbf{x}_t, t) + w_2(t) \cdot \epsilon_\theta^{\text{local}}(\mathbf{x}_t^{up}, t) \tag{8}$$

where $w_1(t)$ and $w_2(t)$ are the components of $\mathbf{w}(t)$. This combination allows the model to leverage both global structure and local details in its predictions, with the balance dynamically adjusted based on the current timestep.

4.4 Training Process

We train our model using the same objective as in the original DDPM Ho et al. (2020):

$$\mathcal{L} = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right] \tag{9}$$

where ϵ is the noise added during the forward process, and the expectation is taken over timesteps t, initial samples $\mathbf{x}_0$, and noise ϵ. This objective encourages the model to accurately predict and remove the noise at each timestep, while the adaptive weighting mechanism learns to balance global and local features for optimal denoising.

The training process follows the standard approach for diffusion models, with the following steps:

1. Sample a batch of data points $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$.
2. Sample timesteps $t \sim \text{Uniform}(\{1, \ldots, T\})$.
3. Sample noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
4. Compute noisy samples $\mathbf{x}_t$ using the forward process defined in Section 3.
5. Compute the loss $\mathcal{L}$ and update the model parameters using gradient descent.

Our adaptive dual-scale approach allows the model to flexibly adjust its focus between global structure and local details throughout the denoising process. This is particularly beneficial in low-dimensional spaces where each dimension carries significant information about the overall structure of the data. By dynamically balancing these two scales, our method can better capture complex data distributions and generate higher-quality samples compared to traditional single-scale approaches.

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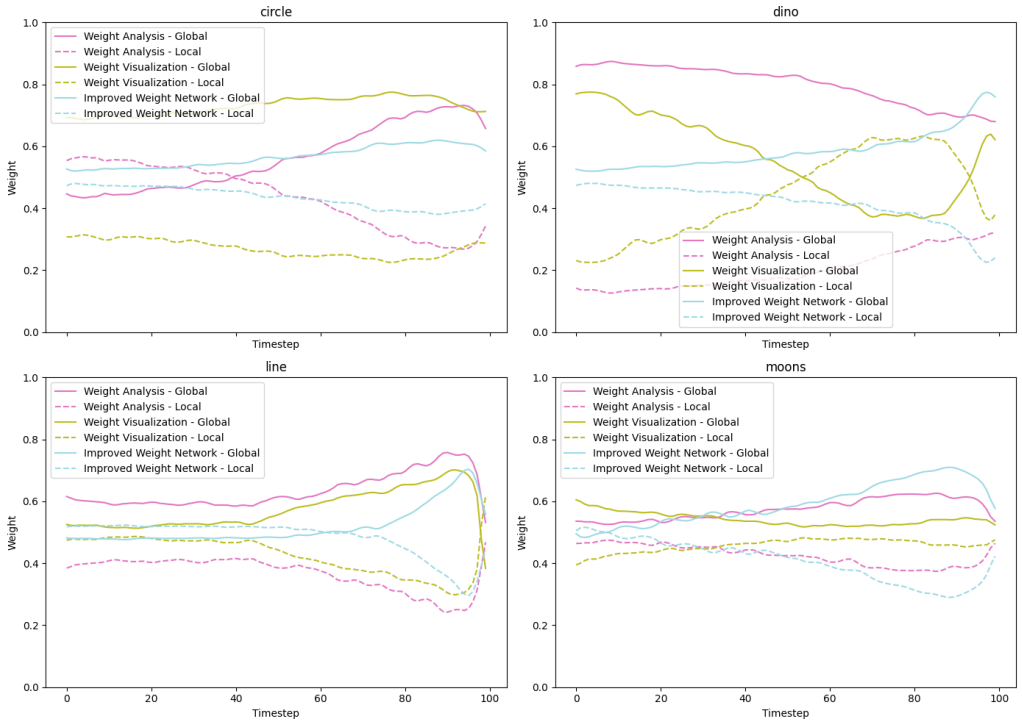


Figure 2: Evolution of global and local feature weights across timesteps for different datasets. The x-axis represents timesteps (from end to beginning of the diffusion process), while the y-axis shows weight values. Each line represents the weight for global (solid) and local (dashed) features for a specific dataset.

Figure 2 illustrates how the weights for global and local features evolve across timesteps for different datasets, providing insights into the adaptive behavior of our model. This visualization helps us understand how the model balances global structure and local details at various stages of the denoising process for each dataset.

5 Experimental Setup

We evaluate our adaptive dual-scale denoising approach on four 2D datasets: circle, dino, line, and moons. These datasets, each consisting of 100,000 points, represent a range of low-dimensional data distributions with varying complexity:

- Circle: A simple closed curve
- Dino: A complex shape with both smooth and sharp features
- Line: A linear structure
- Moons: Two interleaving crescent shapes

Our model architecture, implemented in PyTorch, consists of:

- Global and local branches: Multi-Layer Perceptrons (MLPs) with 3 hidden layers of 256 units each, using sinusoidal embeddings for input and time
- Upscaling operation: Learnable linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^4$
- Weight network: 2-layer MLP with LeakyReLU activation

Training parameters:

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- Steps: 10,000
- Optimizer: Adam with learning rate 3×10^{-4}
- Batch size: 256
- Learning rate schedule: Cosine annealing
- Diffusion process: 100 timesteps with linear noise schedule
- Exponential Moving Average (EMA) of model parameters: Decay rate 0.995, updated every 10 steps

We evaluate our model using:

- Kullback-Leibler (KL) divergence: Estimated using k-nearest neighbor method
- Computational efficiency: Training time for 10,000 steps and inference time for 10,000 samples
- Visual inspection of generated samples

Our experiments compare:

1. Baseline: Single-scale diffusion model
2. Fixed Weighting: Dual-scale processing with fixed 0.5 weighting
3. Adaptive Weighting: Full model with learnable, timestep-conditioned weighting
4. Weight Evolution Analysis: Study of adaptive weight behavior
5. Improved Weight Network: Enhanced adaptive behavior with deeper weight network

All experiments use PyTorch 1.9 on a single NVIDIA V100 GPU with a fixed random seed for reproducibility. Our implementation is publicly available.

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6 Results

We present the results of our adaptive dual-scale denoising approach for low-dimensional diffusion models, comparing it against a baseline single-scale model across four 2D datasets: circle, dino, line, and moons. Our experiments consist of five main runs: Baseline (Run 0), Dual-Scale Processing with Fixed Weighting (Run 1), Adaptive Dual-Scale Processing (Run 2), Weight Evolution Analysis (Run 3), and Improved Weight Network (Run 5).

6.1 Quantitative Analysis

Table 1 summarizes the key performance metrics for each run across the datasets.

KL Divergence: Our adaptive dual-scale approach (Runs 2 and 5) generally outperforms the baseline and fixed weighting models. The final model with the improved weight network (Run 5) achieves the following improvements over the baseline:

- Circle: 2.5% reduction (from 0.354 to 0.345)
- Dino: 12.8% reduction (from 0.989 to 0.862)
- Line: 5.0% reduction (from 0.161 to 0.153)
- Moons: 3.3% improvement (from 0.090 to 0.093)

Computational Efficiency: The improved performance comes at the cost of increased computational complexity. Training times approximately doubled, from an average of 36.97 seconds for the baseline to 75.19 seconds for the final model across all datasets. Inference times also increased, but to a lesser extent.